TALENTRA

Database Design Hackathon

Milestones 1–3: Deliverables & Design Decisions

A working design document based on the Talentra Project Docket

Scope covered

Milestone 1 — Database provider, module inventory and common module assignment

Milestone 2 — Profile and Jobs & Applications entity lists and validation

Milestone 3 — Relationship models and validation for both common modules

Status note: This document provides proposed deliverables for team review. Mark a deliverable approved only after the team has reviewed and accepted it.

How to use this document

This document consolidates the deliverables for the first three milestones in the Talentra docket. The docket defines the required outputs; entity names, cardinalities and implementation choices below are proposed design decisions that the team should validate. No physical SQL tables are created in these milestones.

Milestone overview

Milestone	Backlogs	Required deliverables
1 — Identify and fix DB provider	B1–B3	Provider decision; agreed module list; common module assignment
2 — Entities	B4–B7	Profile entity list and validation; Jobs & Applications entity list and validation
3 — Relationships	B8–B11	Profile relationship model and validation; Jobs & Applications relationship model and validation

Design principles carried forward

- Use PostgreSQL as the selected database engine, subject to team approval and local version verification.
- Keep the three Phase 1 common modules aligned: Master Data, Profile, and Jobs & Applications.
- Use singular snake_case names for proposed table/entity names, consistent with the docket's naming standard.
- Keep entity identification, relationship modelling, column definition, and keys/constraints as separate design steps.
- Treat the relationship cardinalities in this document as proposals to validate, not as requirements explicitly dictated by the docket.

Milestone 1 — Identify and fix the DB provider

Docket reference: page 7, B1–B3.

B1. Database provider decision

Selected provider: PostgreSQL

Reasoning: Talentra requires relational integrity across candidates, jobs and applications; transactional workflows; analytical queries; constraint-based business rules; and performance validation. PostgreSQL provides foreign keys, check constraints, transactions, indexing, partitioning, window functions, common table expressions and materialized views. Its range types and exclusion constraints can support non-overlapping time-window rules such as gig-worker assignments.

Decision record

Field	Decision
Database engine	PostgreSQL
Scope	Common design and implementation for Talentra
Rationale	Integrity, transactional consistency, analytical features and advanced constraints
Validation	Confirm installed version and test required features; measure performance using the agreed synthetic workload

B2. Agreed module inventory

#	Module	Phase / ownership
1	Master data	Phase 1 — whole team
2	Profile	Phase 1 — whole team
3	Jobs and Applications	Phase 1 — whole team
4	Administration	Phase 2 — individual module
5	Organisation and account	Phase 2 — individual module
6	Gig marketplace	Phase 2 — individual module
7	Licences, seats, devices and sessions	Phase 3 — assigned member
8	Sourcing and consultant requests	Phase 3 — assigned member
9	Subscriptions, wallet, payments and download history	Phase 3 — assigned member
10	Notifications, preferences and community	Phase 4 — whole team
11	Analytics layer and reporting marts	Phase 4 — whole team / assigned module

Optional stretch domains (separate from the 11 modules): salary benchmarks, company ratings, trending job snapshots, dashboard aggregates and feature flags.

B3. Common module assignment

Common module	Assignment	Purpose
Master data	Whole team	Shared reference data: skills, qualifications, locations, industries, job roles, work types, languages and enumerations
Profile	Whole team	Candidate identity and professional profile, resumes and profile history
Jobs and Applications	Whole team	Job requirements, questionnaires, applications and recruitment pipeline

Suggested team coordination (not a docket ownership rule)

- Member A: coordinate decisions and maintain the consolidated design document.

- Member B: review Profile requirements and entity coverage.
- Member C: review Jobs & Applications requirements and workflows.
- All members: jointly approve shared entities, relationships and cross-module assumptions.

Milestone 1 acceptance checklist

- B1 — Provider decision and justification recorded.
- B2 — All 11 modules and their scope recorded and agreed.
- B3 — Master Data, Profile, and Jobs & Applications assigned to the whole team.

Milestone 2 — Identify entities

Docket reference: page 7, B4–B7. The lists below are proposed entities derived from the docket's mandatory and expected coverage.

B4. Profile entity list

Entity	Purpose / scope
candidate_profile	Core candidate profile and personal details
candidate_contact	Contact methods and contact details; decide whether multiple contacts/history are needed
candidate_education	Candidate education history
candidate_work_experience	Employment and work-experience history
candidate_skill	Junction entity associating candidates with master skills
candidate_language	Candidate language proficiency records, linked to master languages
candidate_project	Candidate projects, role, duration and outcome
resume	A candidate's resume record
resume_version	Saved resume versions
resume_template	Reusable resume templates
candidate_assessment	Candidate assessment attempts and scores
course_enrolment	Candidate course enrolments
profile_completion_score	Profile completeness score or its history, subject to design choice
candidate_activity_log	Candidate login, profile update and resume activity
candidate_profile_view	History of candidate profile views by employers/recruiters
candidate_job_search_history	Candidate job search history
candidate_application_history	Candidate application history, if not fully derived from applications/status history
candidate_shortlist	Candidate shortlist records, subject to integration with application shortlists
candidate_bookmark	Bookmark records; define whether target is a job, candidate or both
referral	Referral tracking record

B5. Profile entity validation

Validation area	Validation rule / question	Status
Identity and contact	Candidate identity and contact information are covered.	Team review required
Professional profile	Education, experience, skills and languages are represented.	Team review required
Projects and resumes	Projects, resume versions and templates are covered.	Team review required
Development	Assessments, course enrolments and completion scoring are covered.	Team review required
Activity and views	Candidate activity, profile views and search history are covered.	Team review required
Engagement	Shortlists, bookmarks and referrals are covered without duplicating another module's ownership.	Team review required
Master data references	Skills, qualifications, languages, locations, job roles and industries use shared master records where applicable.	Team review required

Profile modelling decisions to resolve

- Decide whether candidate_contact stores multiple methods or whether contact history is needed.
- Define a clear target type for bookmarks, or use separate entities for job bookmarks and candidate bookmarks.

- Decide whether candidate_application_history is stored independently or derived from job_application and application_status_history.
- Define whether profile completion score stores only the current score or historical scores.
- Avoid duplicating shared master data inside candidate profile records.

Milestone 2 — Jobs & Applications entities

B6. Jobs & Applications entity list

Entity	Purpose / scope
job_requirement	Main job posting: employer reference, description, lifecycle status, publication state and consultant openness
job_position	Positions or vacancies associated with a job requirement
job_questionnaire	Questionnaire attached to a job requirement
questionnaire_question	Individual questions within a questionnaire
job_application	Application made by a candidate to a job requirement
application_status_history	History of application status transitions
application_questionnaire_response	Candidate responses to questions associated with an application
application_shortlist	Shortlisting records for an application
interview	Interview schedule and relevant interview details
job_offer	Offer details and status
application_joining	Joining outcome and joining date, where applicable
job_search_history	Candidate job-search activity, if owned here rather than in Profile
job_view_history	History of job views, if required by the final design

B7. Jobs & Applications entity validation

Validation area	Validation rule / question	Status
Job lifecycle	Posting, positions, status, publication state and consultant openness are represented.	Team review required
Questionnaires	Questionnaires, questions and candidate responses are supported.	Team review required
Applications	Candidate applications reference the job requirement and candidate.	Team review required
Pipeline	Shortlisted, interviewed, offered, joined and rejected outcomes are represented.	Team review required
History	Status transitions and candidate search/application history are represented without unnecessary duplication.	Team review required
Cross-module ownership	Candidate identity and employer/organisation data reference their owning modules.	Team review required

Jobs & Applications modelling decisions to resolve

- Decide whether a candidate can submit more than one application to the same job requirement.
- Define how questionnaire responses identify the relevant application and question.
- Decide whether multiple interviews and offer revisions are retained as separate records.
- Keep current application status consistent with application_status_history.
- Avoid duplicate ownership of search history and shortlist records across modules.

Milestone 2 acceptance checklist

- B4 — Profile entity list exists.
- B5 — Profile entity list reviewed and validated by the team.
- B6 — Jobs & Applications entity list exists.
- B7 — Jobs & Applications entity list reviewed and validated by the team.

Milestone 3 — Relationship models

Docket reference: page 7, B8–B11. The relationship cardinalities below are proposed design decisions and must be confirmed by the team.

Cardinality quick reference

- One-to-many (1:N): one parent record can be associated with multiple child records.
- Many-to-many (M:N): multiple records on each side are associated through a junction entity.
- One-to-one (1:1): each side corresponds to at most one record on the other side; enforce with a unique constraint in the physical design.
- Optionality is separate from cardinality and must be specified during column and constraint design.

B8. Profile relationship model

Parent / source	Child / target	Proposed cardinality	Reason / note
candidate_profile	candidate_contact	1:N, conditional	Multiple contact methods/history only if supported by requirements
candidate_profile	candidate_education	1:N	A candidate may have multiple education records
candidate_profile	candidate_work_experience	1:N	A candidate may have multiple experience records
candidate_profile	candidate_project	1:N	A candidate may have multiple projects
candidate_profile	candidate_skill	1:N	Junction rows associate candidates with skills
skill	candidate_skill	1:N	Each junction row references one master skill
candidate_profile	candidate_language	1:N	Language proficiency records per candidate
language	candidate_language	1:N	Each junction/association row references one master language
candidate_profile	resume	1:N	A candidate may have multiple resumes
resume	resume_version	1:N	A resume may have multiple saved versions
resume_template	resume_version	1:N, conditional	If versions retain their template reference
candidate_profile	candidate_assessment	1:N	Assessment attempts/scores per candidate
candidate_profile	course_enrolment	1:N	Course enrolment records per candidate
candidate_profile	candidate_activity_log	1:N	Activity records for a candidate
candidate_profile	candidate_profile_view	1:N as subject	A candidate can be the subject of multiple view events
candidate_profile	candidate_job_search_history	1:N	Search history records per candidate

Additional Profile relationships

- candidate_application_history may reference candidate_profile and/or job_application depending on whether it is stored or derived.
- candidate_shortlist and candidate_bookmark need clear actor, target and ownership definitions before final cardinalities are approved.
- referral should identify the referred candidate or referral subject and the referring actor according to the agreed referral workflow.
- candidate_education may reference qualification and location master data; work experience may reference job role and industry master data.

B9. Profile relationship validation

- Every child record has a defined owner and references the correct parent.
- Master data relationships use valid referenced records.
- Candidate-skill and candidate-language associations are protected from accidental duplicates.
- Resume versions belong to the correct resume; template relationships are explicit.
- Activity and profile-view events identify the relevant subject and actor where applicable.
- Bookmarks, shortlists and referral ownership are unambiguous.

Milestone 3 — Jobs & Applications relationships

B10. Jobs & Applications relationship model

Parent / source	Child / target	Proposed cardinality	Reason / note
job_requirement	job_position	1:N	A requirement may contain multiple positions
job_requirement	job_questionnaire	1:N	A requirement may have multiple questionnaires
job_questionnaire	questionnaire_question	1:N	A questionnaire contains multiple questions
candidate_profile	job_application	1:N	A candidate may submit applications to multiple jobs
job_requirement	job_application	1:N	A job requirement may receive multiple applications
job_application	application_status_history	1:N	An application may have multiple status transitions
job_application	application_questionnaire_response	1:N	An application may have multiple answers
questionnaire_question	application_questionnaire_response	1:N	A question can be answered by multiple applications
job_application	application_shortlist	1:N if history retained	Shortlist actions/events associated with an application
job_application	interview	1:N	An application may involve multiple interviews
job_application	job_offer	1:N if revisions/multiple offers allowed	Define whether offer history is retained
job_application	application_joining	0:1 for one final outcome	Use one final joining outcome unless history is required
candidate_profile	job_search_history	1:N	If search history is owned by this module
job_requirement	job_view_history	1:N	If job view history is implemented

Conceptual application flow

```
candidate_profile 1 → N job_application N ← 1 job_requirement
job_application 1 → N application_status_history
job_application 1 → N interview
job_application 1 → N job_offer (if offer revisions/multiple offers are supported)
```

B11. Jobs & Applications relationship validation

- Every application references exactly one candidate and one job requirement.
- Decide and enforce whether duplicate applications to the same requirement are permitted.
- Status history rows belong to the correct application and are retained as history.
- Questionnaire responses reference valid questions and the correct application; ensure the question belongs to a questionnaire applicable to the job.
- Interviews and offers reference the correct application.
- Joining and rejection outcomes are represented consistently with the application lifecycle.
- Avoid duplicated ownership of search history and shortlist records across modules.

Milestone 3 acceptance checklist

- B8 — Profile relationship model exists.

- B9 — Profile relationships reviewed and validated by the team.
- B10 — Jobs & Applications relationship model exists.
- B11 — Jobs & Applications relationships reviewed and validated by the team.

Panel presentation guide

Milestone 1 — Provider and scope

“We selected PostgreSQL because Talentra requires relational integrity, transactional workflows, analytical queries and constraint-based business rules. We documented the 11 modules and agreed that Master Data, Profile, and Jobs & Applications are common Phase 1 modules.”

Milestone 2 — Entities

“We identified the entities needed to represent candidate profiles and the job recruitment lifecycle. We separated entity identification from relationship and physical schema design so that we can validate each layer systematically.”

Milestone 3 — Relationships

“We modelled one-to-many relationships for profile history and recruitment events, and resolved many-to-many relationships through junction entities such as candidate_skill and job_application. Cardinalities that depend on business policy are explicitly marked for team approval.”

Important open decisions before sign-off

- Confirm PostgreSQL version and required feature availability.
- Approve the final module ownership and team responsibilities.
- Decide the ownership of job-search history, application history, shortlists and bookmarks.
- Confirm whether contact methods, resume versions, interviews and offers retain multiple historical records.
- Decide duplicate application policy and how current status is synchronised with status history.
- Confirm optionality, deletion policy and audit requirements in later design milestones.

Scope boundary

This PDF covers the deliverables for Milestones 1–3 only. It does not claim that the team has approved the design, that SQL tables have been implemented, or that the data model has been performance-tested. Those steps require actual team review and later milestones.

Source basis

Primary source: Talentra: Database Design Hackathon — Project Docket, especially Section 4 (data coverage), Section 5 (data standards), Section 7 (milestones and backlogs), pages 3–7. Proposed entity names and relationship cardinalities in this document are design recommendations based on that scope and must be validated by the team.